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B.S. After
Associate
Degree
(Fall-2021)

PHYS 3701

Electronics-I

Solve the following questions.

- i. Define barrier potential. What happens to the barrier potential when the temperature increases?

Solved in Fall 2021 (i)

When the temperature increases, the barrier potential of a p-n junction diode decreases. As temperature increases, carriers gain energy and move more freely. More electrons and holes are generated. The reduced barrier potential allows more carriers to cross the junction.

- ii. What is light emitting diode? How it works-

MBD page 27, 28, 29.

- iii. Define Peak reverse voltage. What is PRV ratio for full wave bridge rectifier - It is particular voltage when applied to a junction we have zero current.

Suppose V_m is peak voltage and I_m is maximum current and R is load resistance, then

$$I_m = \frac{V_m}{R}$$

The direct current through R due to

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diode D_1 is,

$$I_{dc} = \frac{0.318 V_m}{R}$$

The direct current through R due to diode D_1 is

$$I_{dc} = \frac{0.318 V_m}{R}$$

The direct current is

$$\begin{aligned} I_{dc} &= I_{dc} + I_{dc} \\ &= \frac{0.318 V_m}{R} + \frac{0.318 V_m}{R} \end{aligned}$$

$$I_{dc} = \frac{0.636 V_m}{R}$$

The direct voltage across output resistance R is given by,

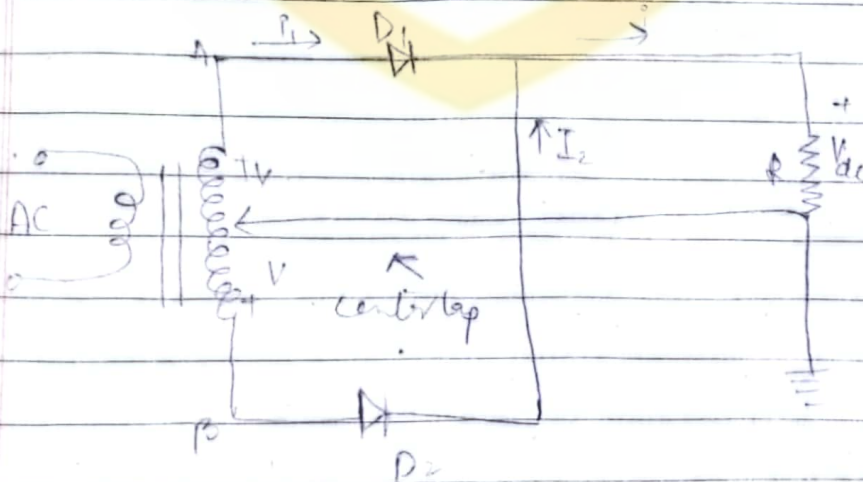
$$V_{dc} = I_{dc} R$$

$$V_{dc} = \frac{0.636 V_m \times R}{R}$$

$$V_{dc} = 0.636 V_m$$

Peak reverse voltage in case of full wave rectifier is,

$$PRV = 2V_m = 2\sqrt{2} V_{rms}$$



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iv. Write down the hybrid equations for two port network circuit- and find out the four hybrid parameters.

We have four variables V_1, I_1, V_2, I_2 .
Consequently if we are to find an equivalent circuit for any network in the box, we shall need four measured parameters. We consider quantities I_1, V_2 as independent variables and I_2, V_1 as dependent variables.
So we can write

$$V_1 = f(I_1, V_2) \quad , \quad I_2 = f(I_1, V_2)$$



Two port network.

Our circuit is operated with ac signals as the effect of changes in terminal quantities can be determined as

$$dV_1 = \left(\frac{\partial V_1}{\partial I_1} \right) dI_1 + \left(\frac{\partial V_1}{\partial V_2} \right) dV_2 \quad \text{and}$$

$$dI_2 = \left(\frac{\partial I_2}{\partial I_1} \right) dI_1 + \left(\frac{\partial I_2}{\partial V_2} \right) dV_2$$

The total derivatives can be replaced by small ac signals

$$dV_1 = v_1, \quad dI_1 = i_1, \quad dV_2 = v_2, \quad dI_2 = i_2$$

So, above equations can be written as

$$V_1 = \left(\frac{\partial V_1}{\partial I_1} \right) i_1 + \left(\frac{\partial V_1}{\partial V_2} \right) v_2 \quad \text{and}$$

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$$i_2 = \left(\frac{\partial I_2}{\partial I_1} \right) i_1 + \left(\frac{\partial I_2}{\partial V_2} \right) V_2$$

$$V_1 = h_{ie} i_1 + h_{re} V_2 \rightarrow \textcircled{a} \text{ and } i_2 = h_{fe} i_1 + h_{oe} V_2 \rightarrow \textcircled{b}$$

Where h_{ie} , h_{re} , h_{fe} and h_{oe} are hybrid parameters.

By short circuiting the output terminals

i.e. $V_2 = 0$ we have

from eq $\textcircled{a}$

$$V_1 = h_{ie} i_1 \Rightarrow h_{ie} = \frac{V_1}{i_1}$$

h_{ie} is called short circuit input impedance.

From eq $\textcircled{b}$

$$i_2 = h_{fe} i_1 \Rightarrow h_{fe} = \frac{i_2}{i_1}$$

h_{fe} is called short circuit forward transfer current ratio

h_{fe} With open circuit at input i.e. $i_1 = 0$, we have

From eq $\textcircled{a}$

$$V_1 = h_{re} V_2 \Rightarrow h_{re} = \frac{V_1}{V_2}$$

h_{re} is called open circuit reverse voltage ratio

From eq $\textcircled{b}$

$$i_2 = h_{oe} V_2 \Rightarrow h_{oe} = \frac{i_2}{V_2}$$

h_{oe} is called open circuit admittance ratio

- v. What is transconductance? Use it to draw the equivalent circuit for the common-emitter transistor amplifier?

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voltage gain for CE amplifier circuit is given by

$$A_{ve} = - \frac{h_{fe} R}{h_{ie}}$$

Where

h_{fe} = forward current ratio
 h_{ie} input resistance

$$\frac{h_{fe}}{h_{ie}} = \frac{I_c / I_B}{V_i / I_B} = \frac{I_c}{V_i}$$

But $V_i = V_{be}$, so

$$\frac{h_{fe}}{h_{ie}} = \frac{I_c}{V_{be}}$$

We define transconductance as

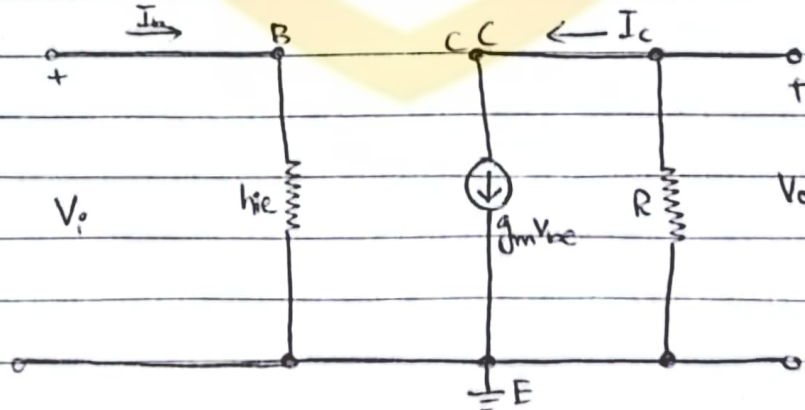
$$g_m = \frac{I_c}{V_{be}}$$

The transconductance may be defined as ratio of output current to input voltage.

We have

$$g_m = \frac{h_{fe}}{h_{ie}} = \frac{I_c}{V_{be}} \Rightarrow I_c = g_m V_{be} = \frac{h_{fe} V_{be}}{h_{ie}}$$

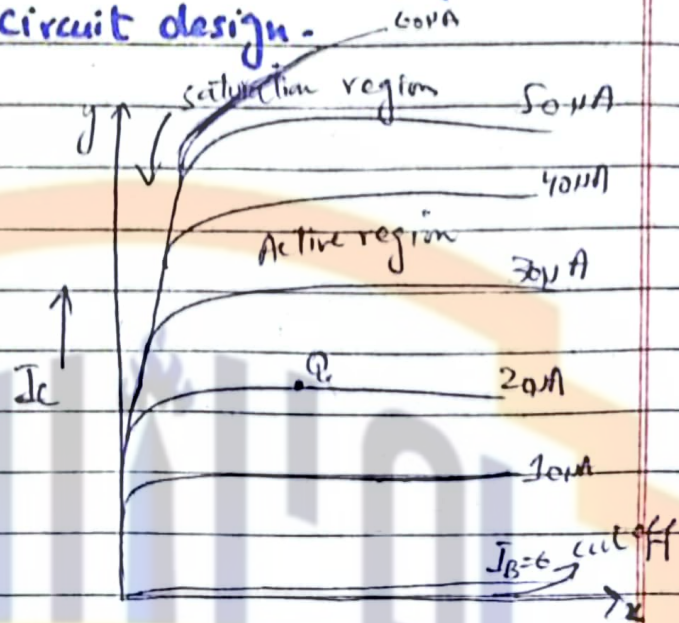
g_m model equivalent circuit is shown in fig



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(vi) Draw the Transistor output characteristic curves and describe the reason behind selection of V_{CE} and I_B as one-half of V_{CE} to start our circuit design.



For a transistor circuit to amplify it must be properly biased with dc voltages. The dc operating point between saturation and cut off is called Q-point. The goal is to set the Q-point such that it does not go into saturation or ~~current~~ cut off when an ac signal is applied.

The reason behind selection of V_{CE} as one-half of V_{CE} is to get maximum swing of ac signal at output.

Long Questions:-

Q#1 (a)

Discuss the circuit operation of a pi-filter when it is connected to a full wave bridge rectifier circuit and determine its ripple factor. (6)

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~~Q#2~~ (b).

Describe the working of positive biased diode clipper by drawing its input and output waveforms. (4)

From Ilmi Book page # 35

Q#2(a)

Draw the circuit diagram of a BJT based common-emitter four resistor bias amplifier circuit and discuss its circuit operation. (6)

From Ilmi book page #109

(b) Draw the circuit diagram of common collector amplifier circuit. Why it is often called emitter follower? (4)

From Ilmi book page # 84.

Q#3 (a)

Explain the construction difference

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between JFET and MOSFET, (6)

From Imiti Book page # 127 & 130

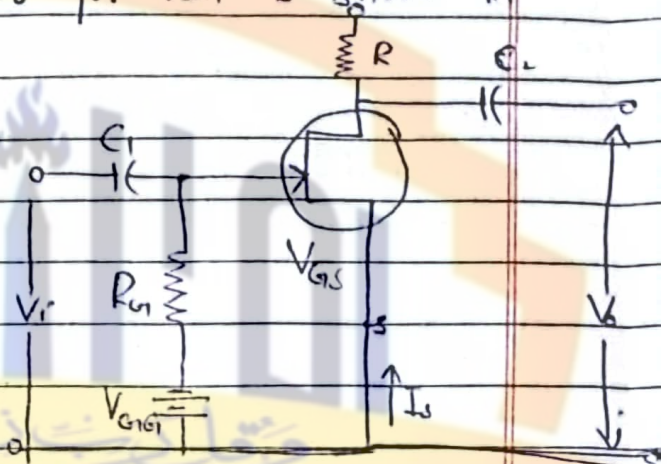
From MBD page # 92 & 96.

(b)

Describe FET based fixed-bias amplifier circuit. (4)

The fixed bias circuit for FET is shown in figure.

We use the Common Source configuration, the C_1 and C_2 are coupling capacitors.



The input loop is given by:

$$V_{GG} = I_D R_G + V_{GS}$$

As I_D current in FET is very small

$$I_D \approx 0, \text{ so}$$

$$I_D R_G \approx 0$$

$$V_G = -V_{GS}$$

This shows that V_{GS} is equal to negative sign of V_{GG} . Since V_{GG} is fixed, so V_{GS} is also fixed.

This circuit is least satisfactory. The fixed biased circuit is not used for minimum and maximum change of transistor parameter also, the disadvantage of this circuit is that the voltage is applied for biasing the or limit the I_D .